

Assignment 1

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Comparison of Verilog Abstraction Levels

Level	Main Focus	Keywords / Constructs	Usage	Synthesizable
Behavioral	Describes what the circuit does	always, initial, if, case	Algorithm description and complex logic	Yes (if written correctly)
Dataflow	Describes data movement using equations	assign, operators	Combinational circuits	Yes
Gate	Describes logic gates directly	and, or, xor, not	Small digital circuits	Yes
Structural	Describes hardware using modules and connections	module instantiation	Hierarchical designs	Yes

1. Gate-Level Full Adder

full_adder_gate.v

```
module full_adder_gate(  
    input A,  
    input B,  
    input Cin,  
    output Sum,  
    output Cout);  
wire x1;  
wire a1;  
wire a2;  
  
xor(x1, A, B);  
xor(Sum, x1, Cin);  
and(a1, A, B);
```

```
and(a2, x1, Cin);  
or(Cout, a1, a2);  
  
endmodule
```

2. Structural Full Adder

full_adder_structural.v

```
module half_adder(  
    input A,  
    input B,  
    output Sum,  
    output Carry);  
  
xor(Sum, A, B);  
and(Carry, A, B);  
  
endmodule  
  
module full_adder_structural(  
    input A,  
    input B,  
    input Cin,  
    output Sum,  
    output Cout);  
    wire s1;  
    wire c1;  
    wire c2;  
    half_adder HA1(  
        A,  
        B,  
        s1,  
        c1);  
  
    half_adder HA2(  
        s1,  
        Cin,  
        Sum,  
        c2);  
  
    or(  
        Cout,
```

```
    c1,  
    c2);  
  
endmodule
```

3. Behavioral Full Adder

full_adder_behavioral.v

```
module full_adder_behavioral(  
    input A,  
    input B,  
    input Cin,  
    output Sum,  
    output Cout);  
assign {Cout, Sum} = A + B + Cin;  
endmodule
```

Testbench

tb.v

```
module tb;
reg A;
reg B;
reg Cin;

wire Sum_gate;
wire Cout_gate;

wire Sum_struct;
wire Cout_struct;

wire Sum_beh;
wire Cout_beh;

full_adder_gate G1(
    A,
    B,
    Cin,
    Sum_gate,
    Cout_gate);
full_adder_structural S1(
    A,
    B,
    Cin,
    Sum_struct,
    Cout_struct);
full_adder_behavioral B1(
    A,
    B,
    Cin,
    Sum_beh,
    Cout_beh);
initial begin

A=0; B=0; Cin=0;
#10;

A=0; B=0; Cin=1;
#10;

A=0; B=1; Cin=0;
#10;
```

```
A=0; B=1; Cin=1;
#10;

A=1; B=0; Cin=0;
#10;

A=1; B=0; Cin=1;
#10;

A=1; B=1; Cin=0;
#10;

A=1; B=1; Cin=1;
#10;

$finish;
end

initial begin

$monitor("A=%b B=%b Cin=%b | Gate=%b%b Structural=%b%b Behavioral=%b%b",
A,B,Cin,Cout_gate,Sum_gate,Cout_struct,Sum_struct,Cout_beh,Sum_beh);
end
endmodule
```